

TrustDesk – AI Support Operations Agent

Capstone One-Pager

Problem Statement

The idea is to build an AI-powered support system that can help support teams handle customer tickets coming from different helpdesk platforms. TrustDesk will understand the issue, use the customer and order context along with company policies, prepare a response, and help the support agent decide the next action.

Implementation Approach

- **Helpdesk Integration** – Integrate with Zendesk, Freshdesk and Intercom. I will keep each integration separate so that the same TrustDesk workflow can work with different support platforms.
- **Webhooks** – Set up webhooks for ticket creation and updates. When a new ticket comes in or an existing ticket changes, the helpdesk will send an event to TrustDesk and start the processing flow.
- **Ticket and Customer Context** – After receiving the webhook, fetch the ticket conversation and the required customer and order details. Store the required information so the support agent can see the complete context.
- **Knowledge Base and RAG** – Load company support documents and policies into a knowledge base. Use embeddings and retrieval to find the relevant information for each ticket and pass that context to the LLM.
- **AI Ticket Triage** – Classify the ticket into areas such as shipping, refund, warranty, billing, account security or general. Also assign a priority and decide if the ticket needs to be escalated.
- **Draft Reply with Citations** – Generate a response using the ticket information and retrieved policies. The draft will include the relevant knowledge-base references. If the available information is not enough, TrustDesk should escalate instead of guessing.
- **Actions with Human Approval** – For actions such as refund review or replacement orders, the AI will recommend the action but a support agent must approve it before execution. Idempotency will be used so the same action is not performed twice.
- **Guardrails** – Add checks for prompt injection, requests for hidden information, unsupported policy claims, PII leakage and unauthorized actions. Security checks will be handled by the application and not only by the AI prompt.
- **Tracing and Evaluation** – Keep a basic trace of each AI run, including the ticket, retrieved documents, actions and final result. Use the provided evaluation cases to check triage accuracy, citation coverage, escalation and unsafe-action blocking.

Overall Flow

Zendesk / Freshdesk / Intercom → Webhook → TrustDesk → Ticket + Customer/Order Context → Knowledge Retrieval → AI Triage → Draft Response + Citations → Human Review → Action
